

Agney Praseed

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Education

Brandenburg University of Technology (BTU), Cottbus-Senftenberg, Germany
Master of Science in Artificial Intelligence

Apr 2025 – Present

National Institute of Technology Puducherry, India
Bachelor of Technology in Computer Science and Engineering
CGPA: 7.73/10

May 2022

Professional Experience

Publicis Sapient, Bengaluru, India
Software Development Engineer

June 2022 – February 2024

- **Built and maintained a full-stack enterprise assessment platform. Used Java/Spring Boot backend with PostgreSQL, Angular frontend, deployed on Azure.**
- **Owned the API layer end-to-end: designed RESTful endpoints, optimized performance bottleneck, and ensured scalability under production load.**
- **Wrote unit and integration tests and contributed to NFR compliance in an agile team.**

Technical Skills

Programming Languages: Java, C++, Python, JavaScript, LaTeX, Bash, SQL

Databases: PostgreSQL, MySQL, MongoDB

Tools: Git, Docker, AWS, Azure, JMeter, JIRA, Confluence, Elasticsearch

Libraries/Frameworks: ReactJS, Redux, Scikit-learn, pandas, Hugging Face Transformers, Spring Boot, Node.js, Express.js

Projects

Travel Aggregator — Cloud-Native Travel Booking Platform

- Architected and developed 3 independently deployable microservices (Search, Bookings, Payment Gateway) utilizing Spring Boot, Node.js/Koa, and MySQL on AWS RDS.
- Built a responsive React 17 SPA with 20+ routes and complex state management using React Context API, eliminating the need for Redux.
- Engineered an AWS OpenSearch gateway via Spring Boot, implementing RESTful endpoints and frontend DSL queries to serve flights, accommodations, and packages. Designed relational data models with JPA/Hibernate across 5 complex entities.
- Integrated Keycloak for OAuth 2.0/JWT-based authentication and developed a custom Node.js payment gateway supporting OTP generation, MFA, and multi-method payment flows (Credit, UPI, EMI).
- Integrated Google Maps API for real-time geolocation, place autocomplete, and distance/duration calculations for cab bookings.
- Dockerized the full stack, configured automated Jenkins CI/CD pipelines (SonarQube, Cobertura, Cucumber BDD), and performed JMeter load testing to ensure stability for up to 3,000 concurrent users.

Preference-Aligned Chatbot with Gemma & DPO

- Fine-tuned Google's Gemma-2B language model using Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) on Anthropic's HH-RLHF dataset
- Applied LoRA for parameter-efficient training, optimizing only 0.02% of model parameters.

- Built a complete alignment pipeline: data preprocessing → SFT → DPO → deployment
- Deployed trained model adapters to HuggingFace Hub and built a live chatbot
- Tech: Python, PyTorch, HuggingFace Transformers, TRL, PEFT, LoRA

Car Detection for Autonomous Driving

- Built an end-to-end object detection pipeline to localize cars in road images using the YOLOv2 architecture and pretrained weights.
- Built and Optimized anchor box mechanisms and grid-based prediction system to detect multiple objects per image with precise bounding box coordinates and confidence scores
- Technologies: Python, TensorFlow, Keras, YOLO, OpenCV, NumPy, Transfer Learning, Object Detection

Certificates

Microsoft Certified: Azure Data Scientist Associate

February 2026

AWS Certified Cloud Practitioner

September 2022

Neural Networks and Deep Learning (Coursera)

Convolutional Neural Networks (Coursera)

Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization (Coursera)

Sequence Models (Coursera)

Fundamentals Of Deep Learning (NVIDIA DLI)

AWS Machine Learning Foundations

Elements of AI (University of Helsinki)

AI For Everyone (Coursera)

The Bits and Bytes of Computer Networking (Coursera)

Full Stack Open (University of Helsinki)

Java Programming I (University of Helsinki)

Algorithms (Robert Sedgewick)

Languages

- **English:** C1 - Advanced (Full Professional Proficiency)
- **German:** A2

Interests

- Football (viewing and competing) • Deep Learning Enthusiast